

Supply Pulse Logistics | SQL + Power BI Portfolio Project

Project Overview

Developed an end-to-end business intelligence solution using SQL and Power BI for a logistics and supply chain company. The project focused on analyzing sales performance, shipment operations, inventory efficiency, profitability trends, and warehouse performance. SQL was used for data analysis, aggregations, joins, ranking functions, and business logic creation, while Power BI was used to design interactive dashboards and KPI reporting.

Tools & Technologies

- SQL (PostgreSQL)
- Power BI
- DAX Measures
- Power Query
- Data Modeling
- Window Functions & CTEs
- Business Intelligence & KPI Reporting

Core KPIs

Metric	Value
Total Sales	631.68M
Total Orders	15K
Average Order Value	42.11K
Total Profit	285.30M
Inventory Turnover Ratio	501.36
Low Stock Items	237
Total Shipments	10K
Average Delay Days	1.88
MoM Growth %	4.47%

Dashboard Structure

- Executive Overview Dashboard
 - Sales performance, order trends, profitability, delivery status analysis
- Inventory Analysis Dashboard
 - Stock levels, low-stock risk, aging inventory, warehouse performance
- Logistics & Shipment Dashboard
 - Shipment mode analysis, carrier performance, delay tracking, carrier cost analysis
- Sales Trend Dashboard
 - Month-over-month growth, quarterly sales trends, YoY growth analysis

Key SQL Queries Used

1. Profit Margin % by Category

```
SELECT p.category,  
SUM((p.selling_price-p.unit_cost)*o.quantity_ordered) /  
SUM(o.sales_amount)*100 AS profit_margin_percent  
FROM fact_orders o  
INNER JOIN dim_products p  
ON o.product_id=p.product_id  
GROUP BY p.category  
ORDER BY profit_margin_percent DESC;
```

2. Top 3 Products by Sales using CTE

```
WITH cte_ranking AS(  
SELECT p.product_name,  
SUM(o.sales_amount) AS total_sales,  
RANK() OVER(ORDER BY SUM(o.sales_amount) DESC) AS ranking  
FROM fact_orders o  
INNER JOIN dim_products p  
ON o.product_id=p.product_id  
GROUP BY p.product_name  
)  
SELECT * FROM cte_ranking  
WHERE ranking <=3;
```

3. Running Total Sales by Month

```
SELECT DATE_TRUNC('month',order_date) AS months,  
SUM(sales_amount) AS monthly_sales,  
SUM(SUM(sales_amount)) OVER(  
ORDER BY DATE_TRUNC('month',order_date)  
) AS running_total_sales  
FROM fact_orders  
GROUP BY DATE_TRUNC('month',order_date)  
ORDER BY months;
```

4. Inventory Risk Ranking

```
SELECT product_id,  
ROUND(AVG(inventory_age_days),2) AS avg_inv_days,  
ROW_NUMBER() OVER(  
ORDER BY AVG(inventory_age_days) DESC  
) AS ranking  
FROM fact_inventory  
GROUP BY product_id;
```

Key Business Insights

- Total sales exceeded **631M** with over **15K orders**, indicating strong overall business performance.
- Electronics generated the highest revenue contribution at approximately **151M**, making it the strongest-performing category by sales.
- Industrial products achieved the highest profit margin at nearly **49%**, highlighting stronger profitability compared to other categories.
- Delivered orders represented nearly **50% of total order volume**, reflecting stable logistics execution and fulfillment operations.
- Ground shipment mode handled the majority of shipments and contributed the highest carrier cost across all shipment methods.
- Inventory analysis identified aging stock above **90 days**, increasing inventory holding costs and operational risk.
- Apparel and Electronics categories showed higher low-stock risk frequency, indicating demand volatility and replenishment challenges.
- Carrier analysis revealed that Canpar and Canada Post experienced the highest average shipment delays among carriers.
- Sales trends showed stronger business performance during May and December, indicating seasonal demand peaks.
- Shipment and inventory optimization opportunities were identified to reduce operational costs and improve fulfillment efficiency.

Business Recommendations

- Implement automated low-stock alerts for high-demand categories such as Electronics and Apparel.
- Reduce aging inventory through inventory clearance strategies and optimized warehouse allocation.
- Improve carrier allocation strategy to reduce shipment delays and optimize logistics costs.
- Focus marketing and sales efforts on high-margin Industrial products to maximize profitability.
- Improve shipment planning to reduce dependency on expensive shipment modes and improve operational efficiency.
- Monitor seasonal sales patterns to improve inventory forecasting and demand planning.

Project Outcome

Successfully built an interactive SQL + Power BI analytics solution that centralized reporting across sales, inventory, shipment, and profitability operations. The project enabled business users to monitor KPIs, identify operational risks, track high-performing categories, and improve data-driven decision-making. This project demonstrates practical experience in SQL analytics, Power BI dashboard development, DAX calculations, business intelligence reporting, and end-to-end data analysis workflows suitable for Data Analyst and Business Intelligence roles.

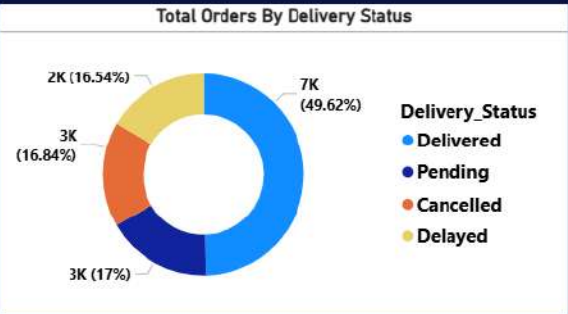
Supply Pulse Logistics | Executive Overview

631.68M	15K	42.11K	45.16	285.30M
Total sales	Total orders	Avg. Order Value	Profit Margin%	Total Profit

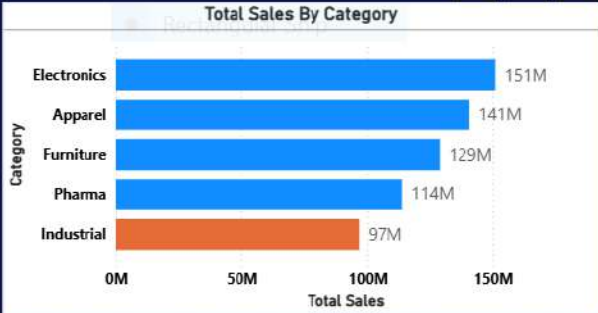
YEAR		
Select all	2023	2024

Key Insight:

- Electronics remains the top revenue-driving category with 151M in sales, while Industrial leads profitability with a 49% profit margin.
- Delivered orders contributed nearly 50% of total order volume, reflecting consistent fulfillment performance.



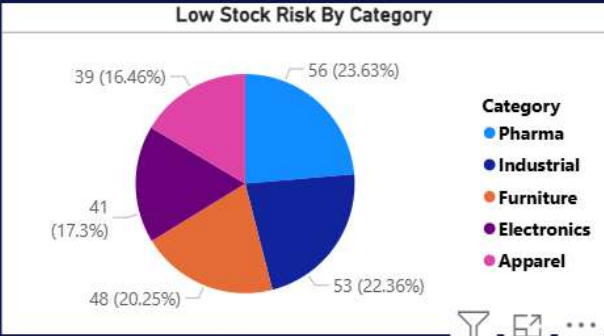
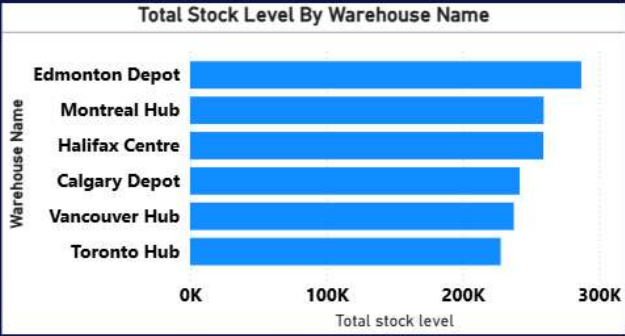
Category		
Apparel	Furniture	Pharma
Electron...	Industrial	



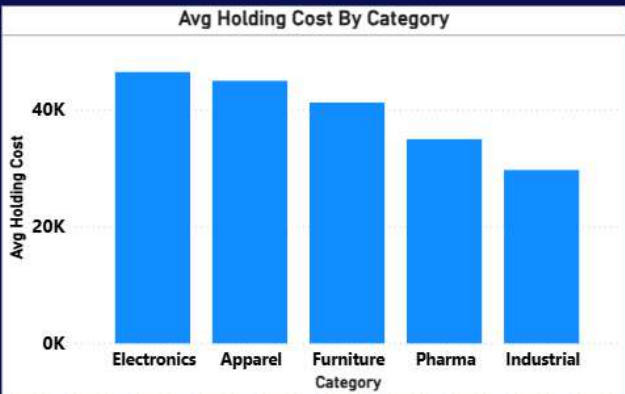
Inventory Analysis

2M	237	39.26K	501.36
Total Inventory	Low Stock Items	Avg Holding Cost	Inventory Turnover Ratio

Key Insight:
Low-stock risk is highest in Electronics and Apparel, while several products maintain an average inventory age above 90 days, increasing holding costs and inventory risk.



Warehouse ID
W001
W002
W003
W004
W005
W006



Top 10 Aging Inventory Products		
Product name	Average of Inventory Age Days	
Sanitizer 5	Rectangular Sni	98.91
Sanitizer 45		87.76
Sanitizer 40		89.81
Gloves 8		95.37
Gloves 43		84.15
Gloves 28		86.30
Generator 49		92.37
Generator 4		79.51
Generator 34		73.00
Generator 29		89.09
Desk 7		81.52



Sales Trend Analysis

Key Insight:
Year-over-year sales decreased marginally by 0.19%, reflecting stable overall business performance despite minor quarterly demand fluctuations.

Category

Apparel

Electronics

Furniture

631.68M
Total sales

4.47
MoM growth %

99.81
YoY growth%

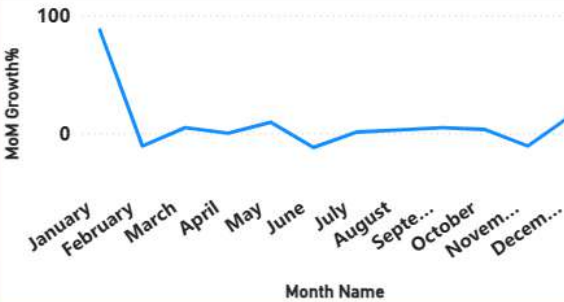
285.30M
Total Profit

year

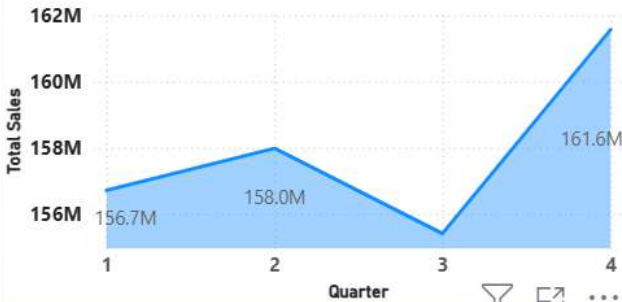
2023

2024

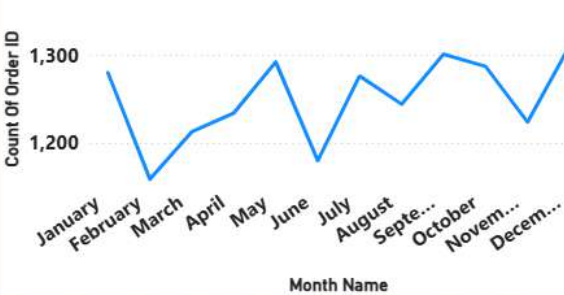
MoM Growth % By Month Name



Total Sales By Quarter



Count Of Order ID By Month Name



YoY Growth% By Year



